

NOVEMBER 2025

The Oyster MSX Project

A Progress Update and Vision
for AVC Diagnostic Services



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The Economic Engine

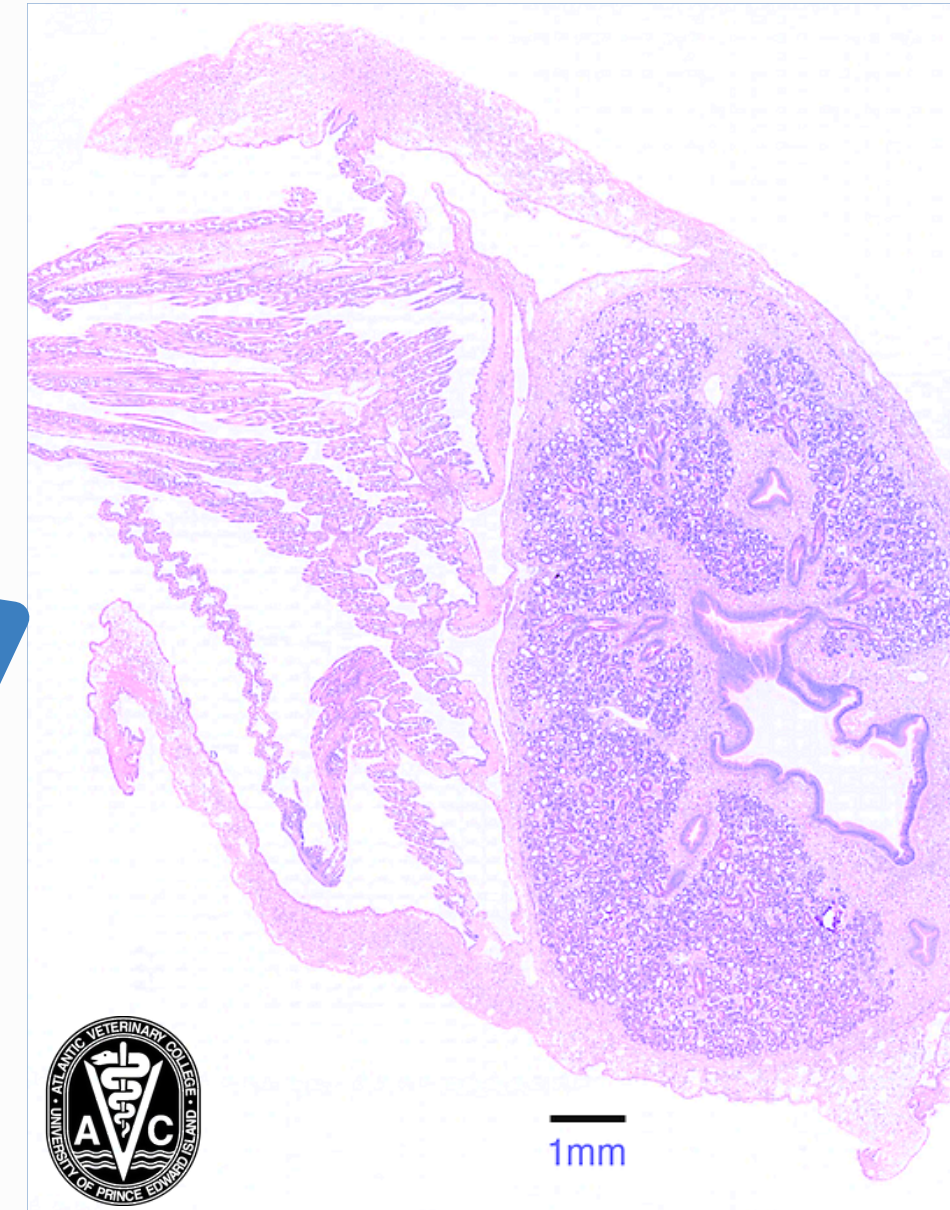
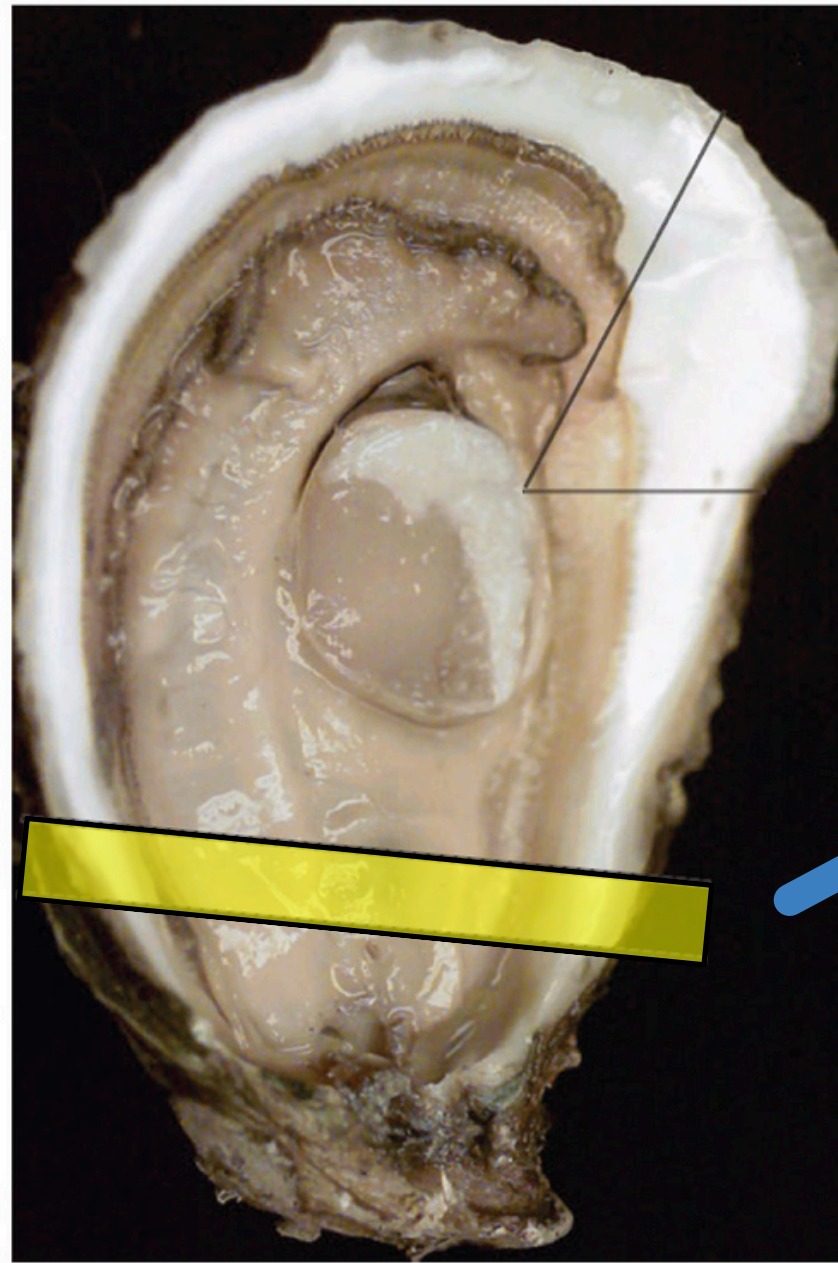
PEI's Oyster Aquaculture Industry

- A Key economic driver for the province, valued at over **\$24 million** annually.
- Supports hundreds of jobs in rural and coastal communities.



From Oyster to Image

Creating the Histology Slide

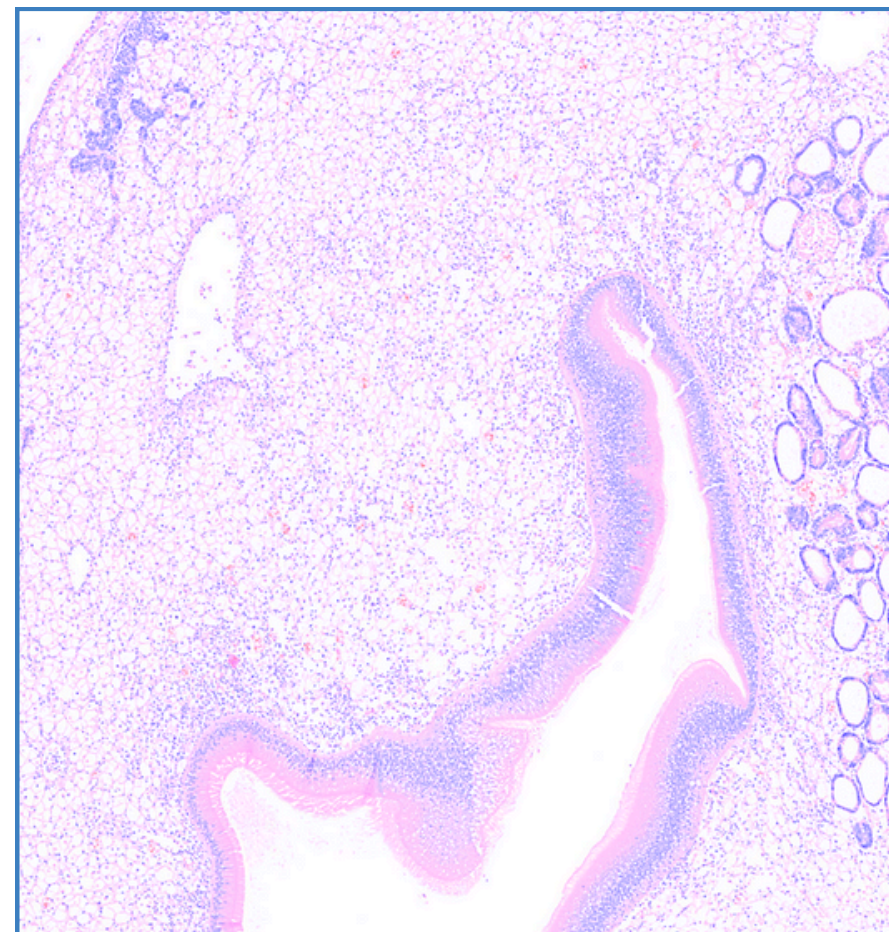
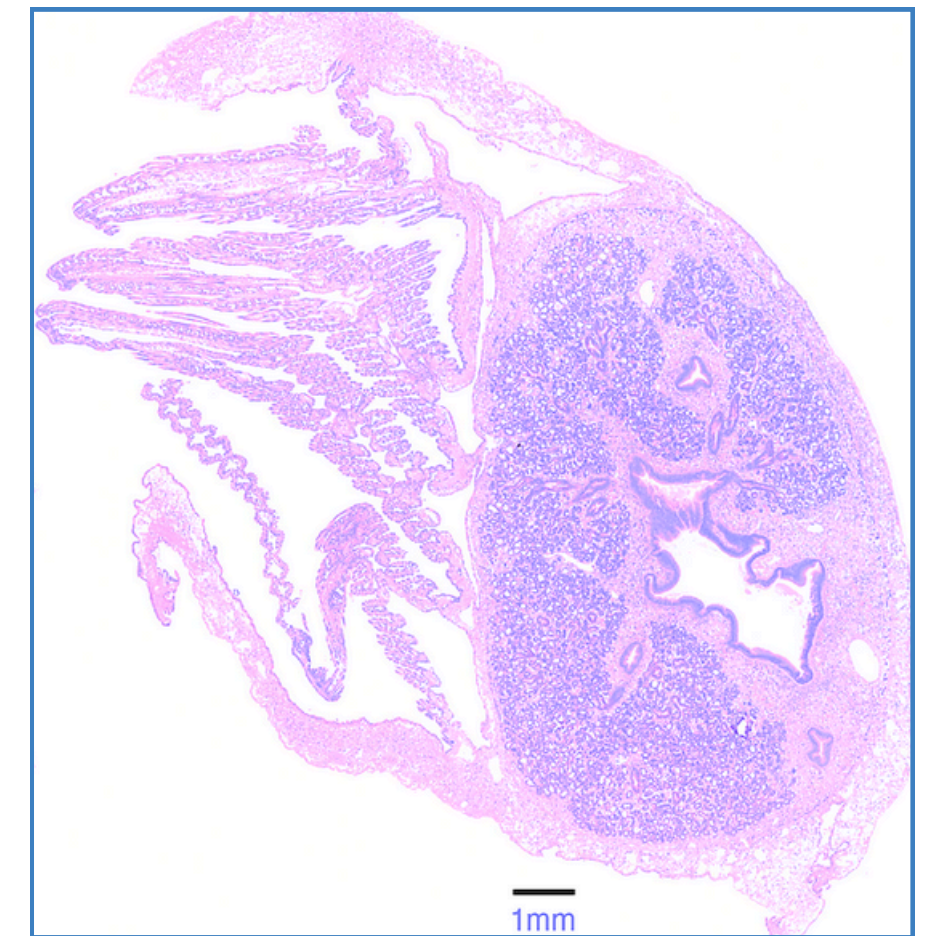


A **thin cross-section** of the entire oyster is processed and stained to make cellular structures visible under a microscope.

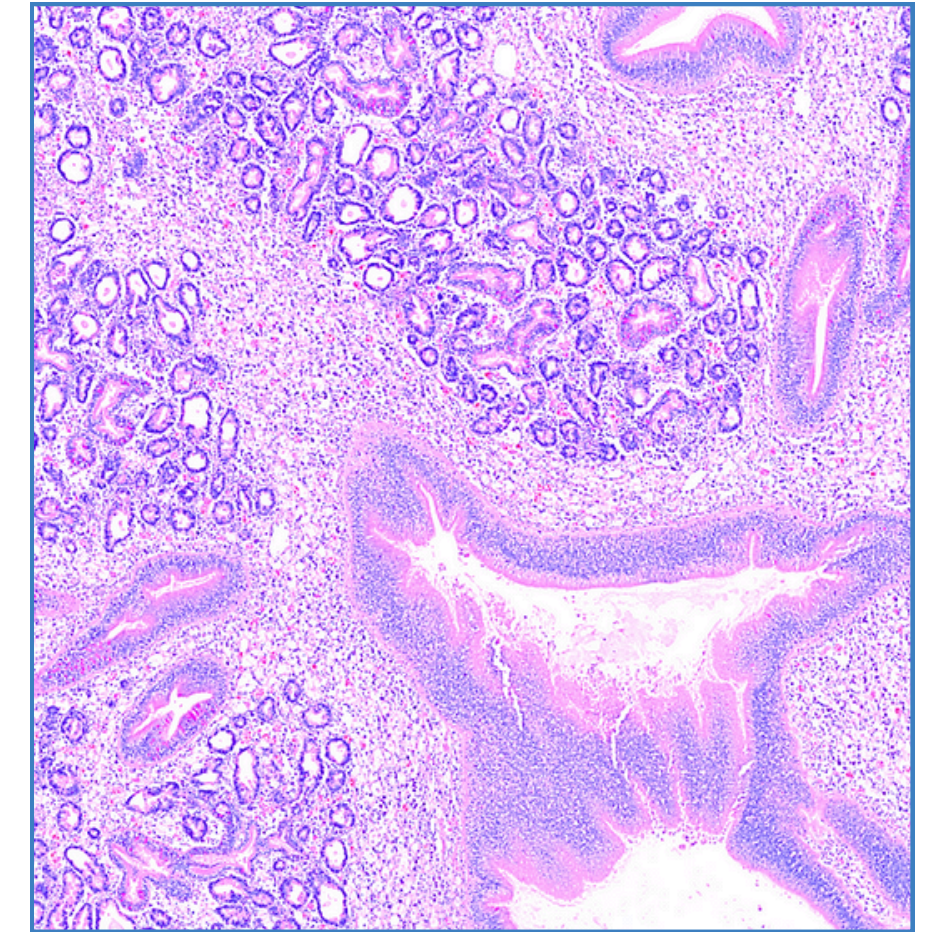
A Microscopic Threat

The MSX Parasite (*Haplosporidium Nelsoni*)

- A protozoan parasite that causes Multinucleated Sphere X (MSX) disease.
- Leads to severe tissue damage and high mortality rates in infected oysters.
- Early and accurate detection is the only way to manage its spread and protect the industry.



Healthy Oyster Tissue

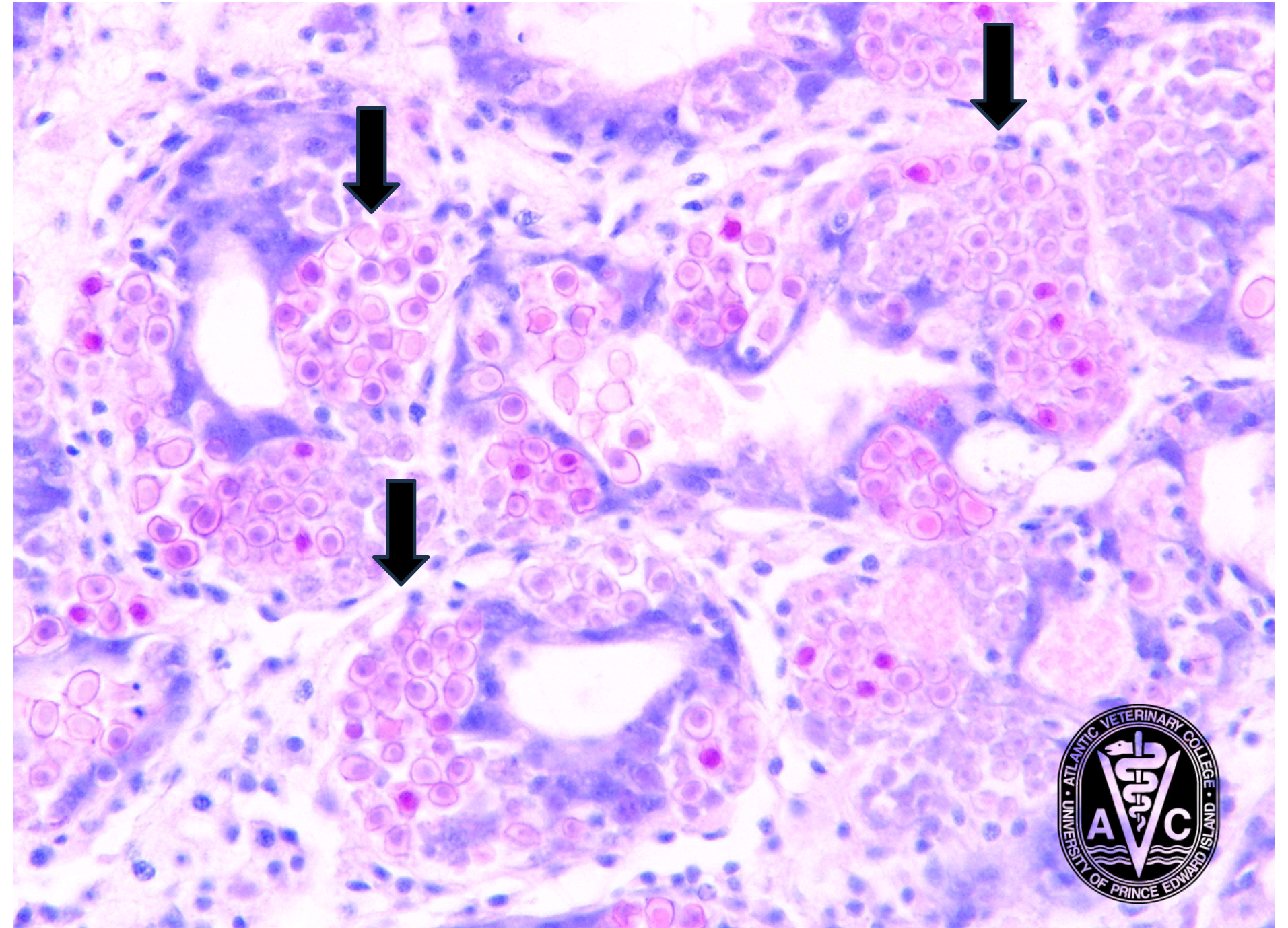


MSX-Infected Tissue

A Microscopic Threat

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A high-magnification view showing multiple **sporocyst clusters** (arrows) of the MSX parasite, surrounded by the oyster's intense **inflammatory response**.

The Diagnostic Front Line

The Role of the Atlantic Veterinary College (AVC)

- The AVC is the primary centre for histology-based diagnosis in Atlantic Canada.
- Histology is the gold standard for assessing disease severity, unlike other methods (e.g., PCR).
- Diagnosis relies on manual slide review by a small number of trained pathologists.
- This process is time-consuming, costly, and creates a significant workflow bottleneck.



The Data Challenge

Building a High-Quality Dataset from Scratch



The Paradox: To build a tool that saves expert time, we first need a large investment of that same limited expert time.

The Strategy

A Human-in-the-Loop
Workflow for Pathology

Stage 1: AI Pre-Analysis



The pipeline performs a
first-pass analysis,
generating initial
annotations on a new slide.



Stage 2: Expert Review



A pathologist quickly
validates and corrects the
AI's suggestions, saving
valuable time.



Stage 3: Model Retraining



The validated annotations
are used to improve the AI
model, making it smarter
with each cycle.

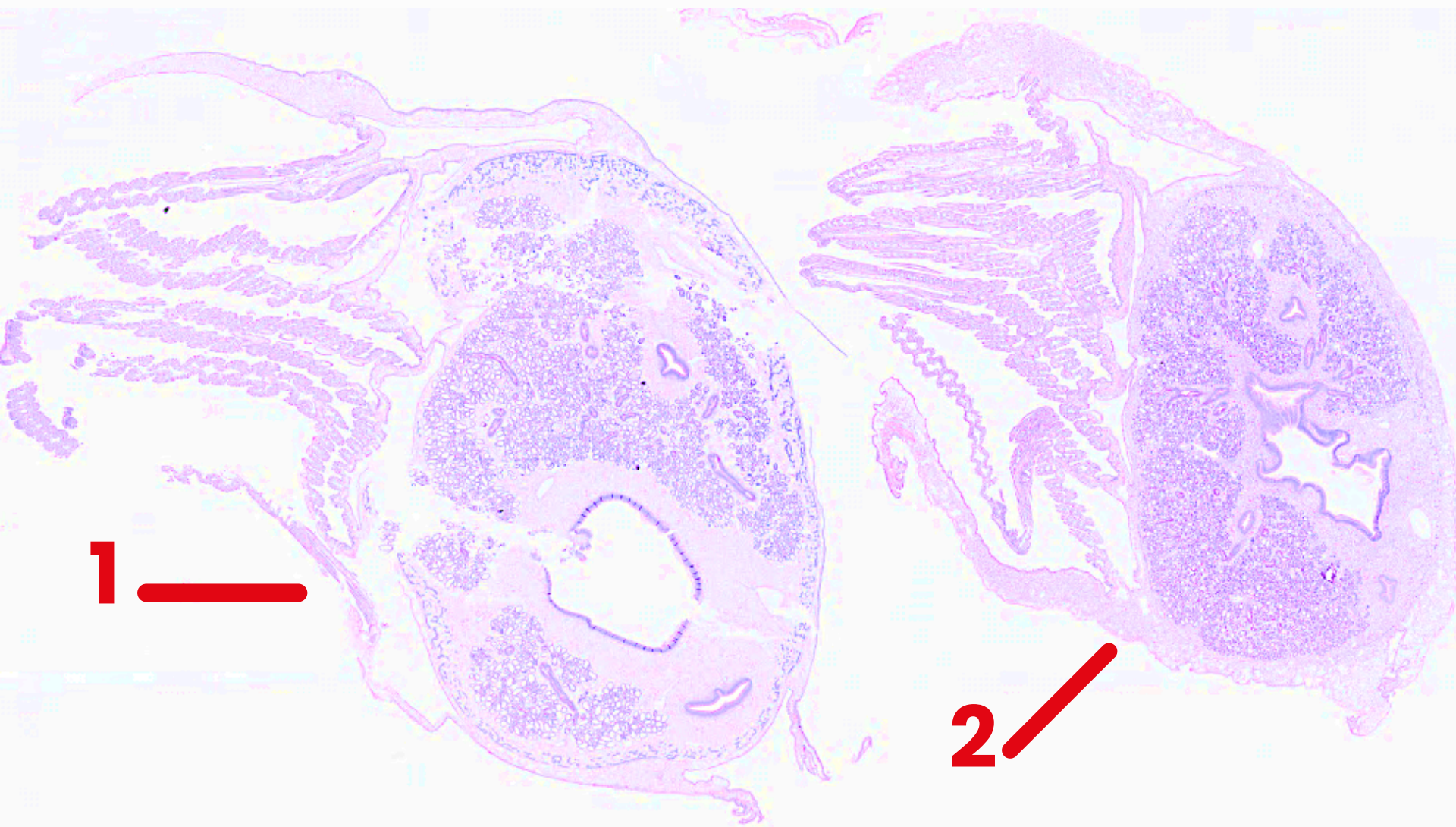


The First Hurdle

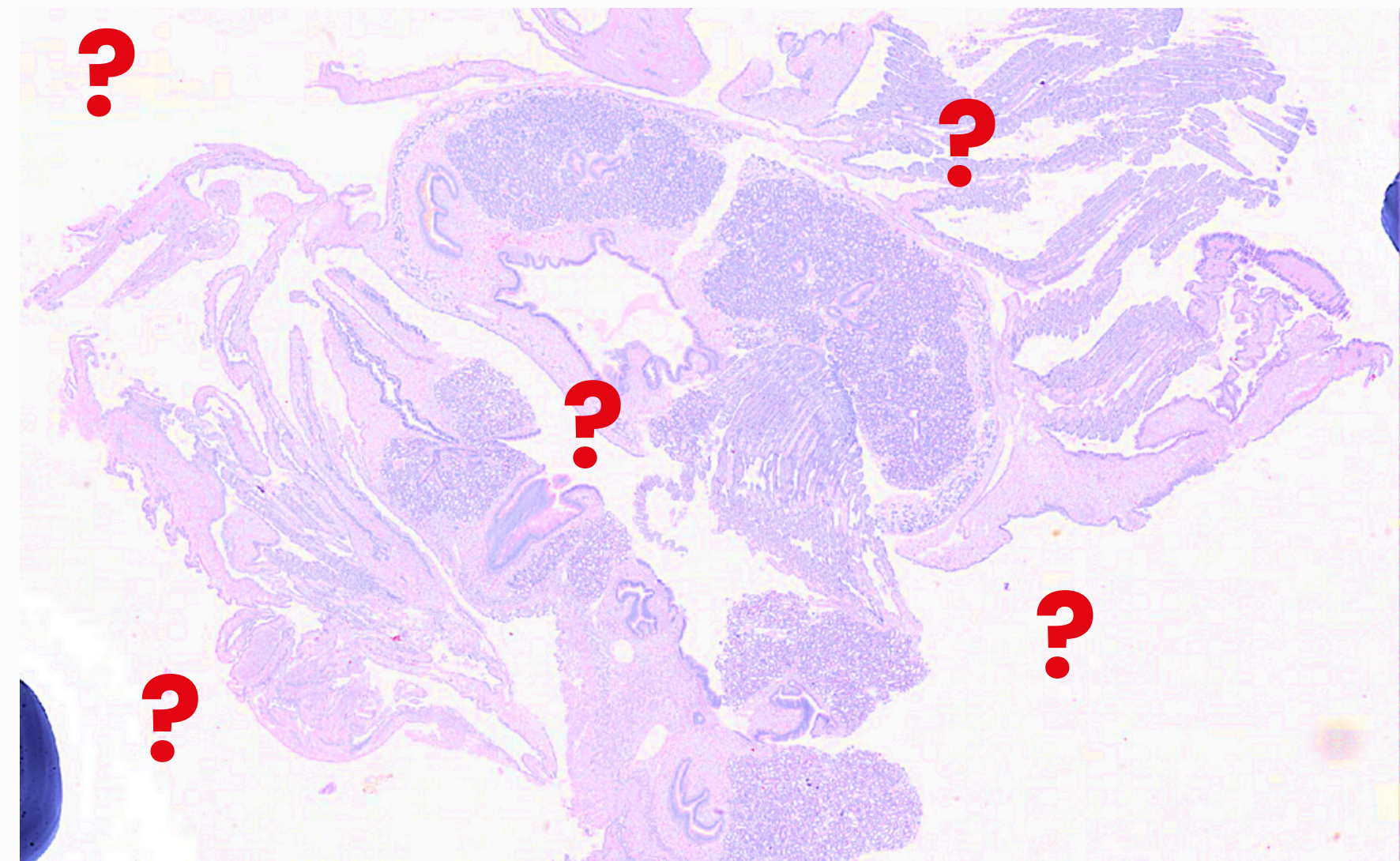
Robustly Separating Oyster Sections

Challenge: How can we create a single, automated process that works reliably for both of these cases without manual intervention?

The Ideal Case



The Common Reality



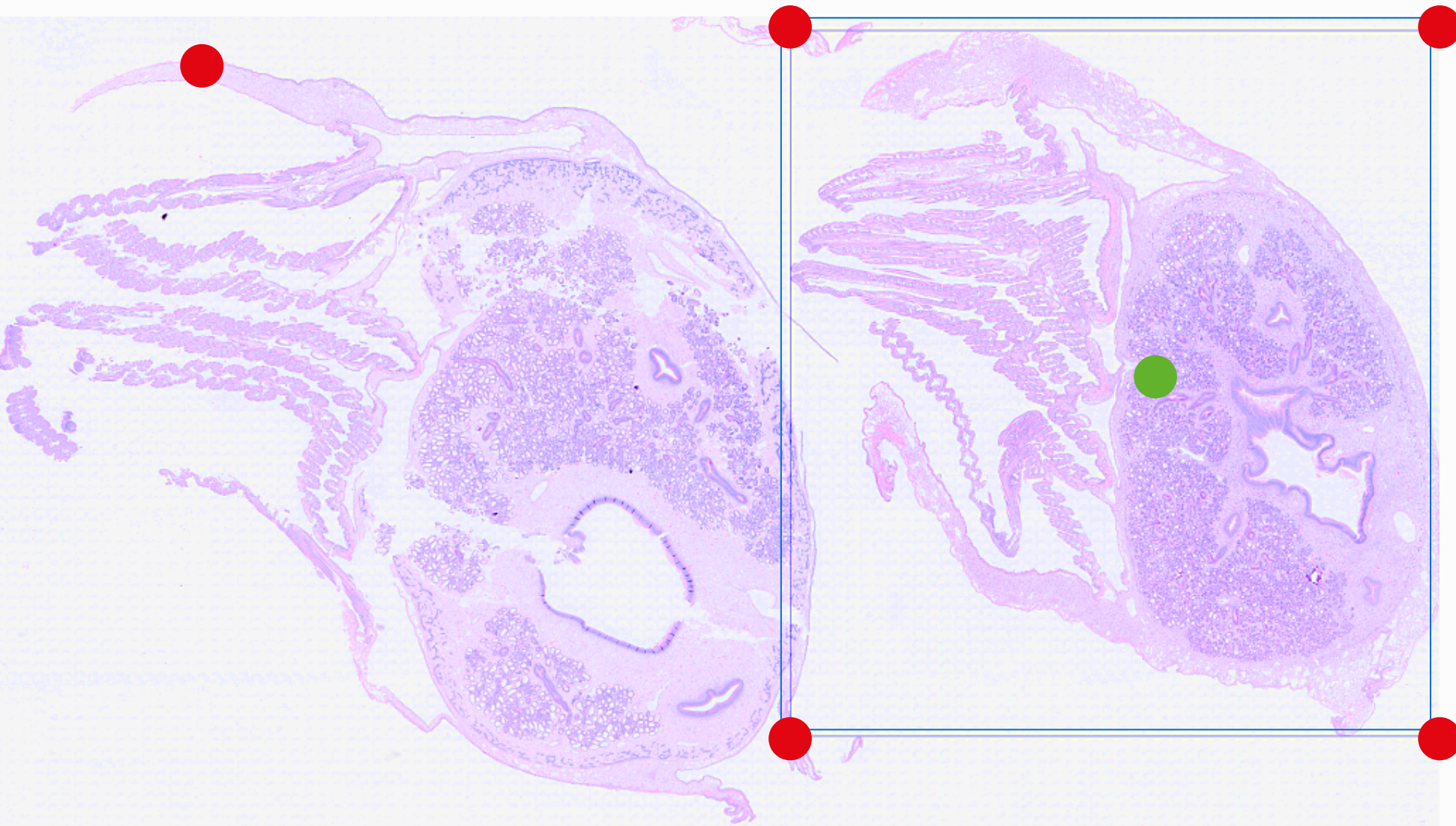
The Solution

A Hybrid Pipeline for Intelligent Prompting

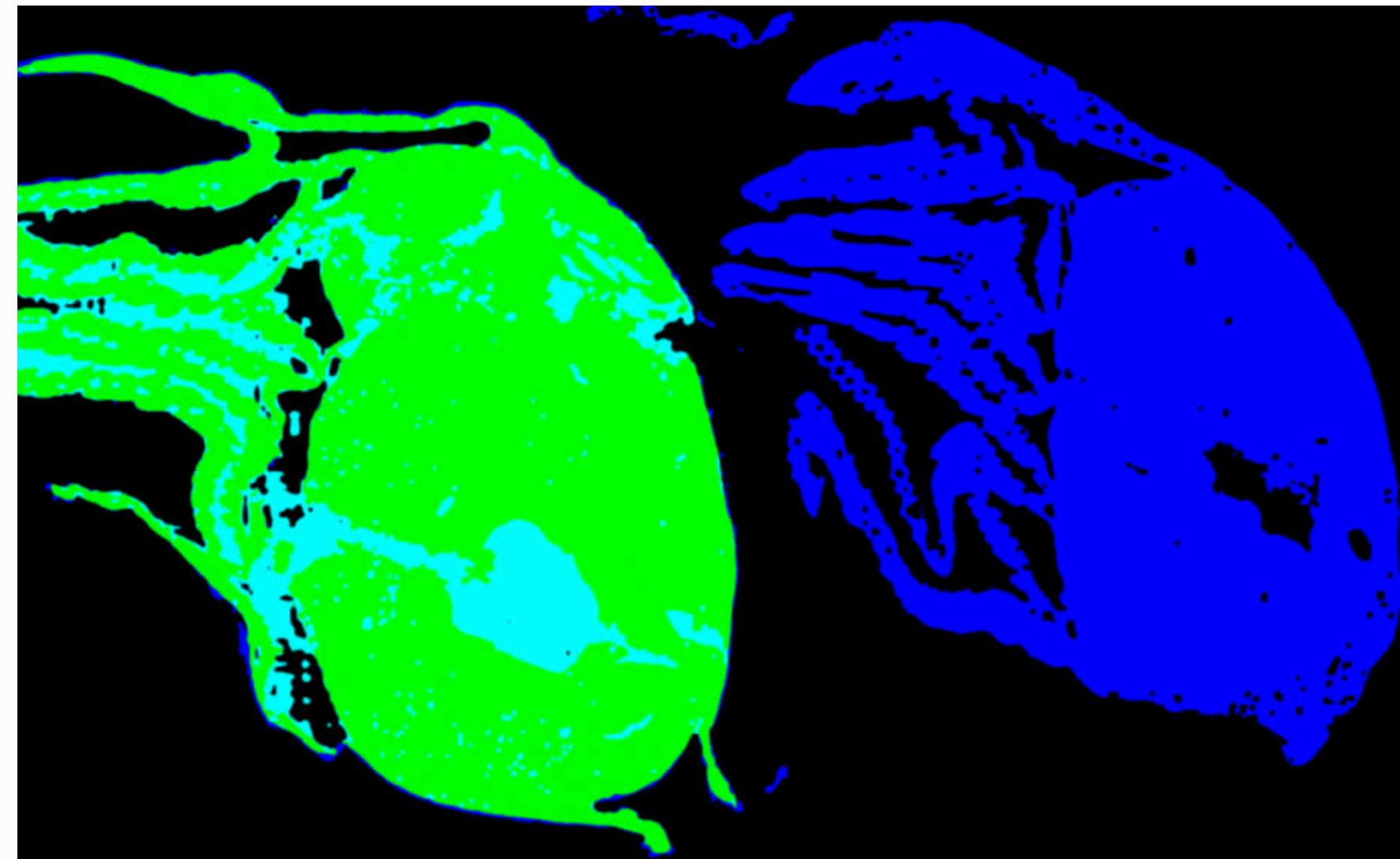
This hybrid strategy uses a simple, reliable method to answer 'where,' and the powerful foundation model to answer 'what.'

- SAM's Guess
- Expert Annotation

Step 1: Generate Robust Prompts (CV)

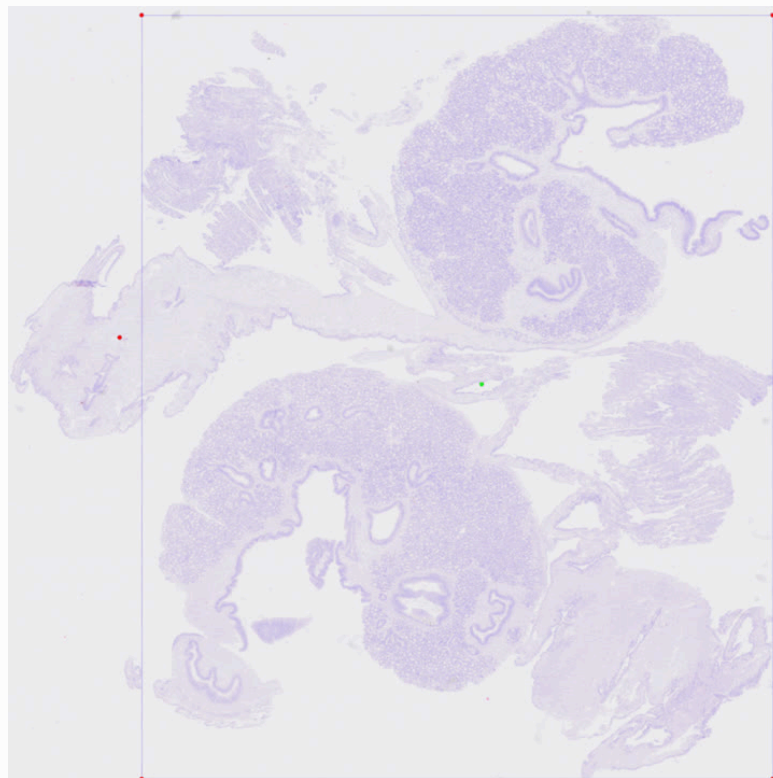


Step 2: Precise Segmentation (SAM)



Current Limitations & Diagnosis

Brittle Prompt Generation



The current CV-based prompter is the weakest link and can fail on complex slides.

Small Dataset Size



While 30 slides generate hundreds of thousands of training patches, they don't capture the full range of biological variability needed for a production-ready model.

These limitations are not roadblocks; they are a clear and actionable guide for what to do next.

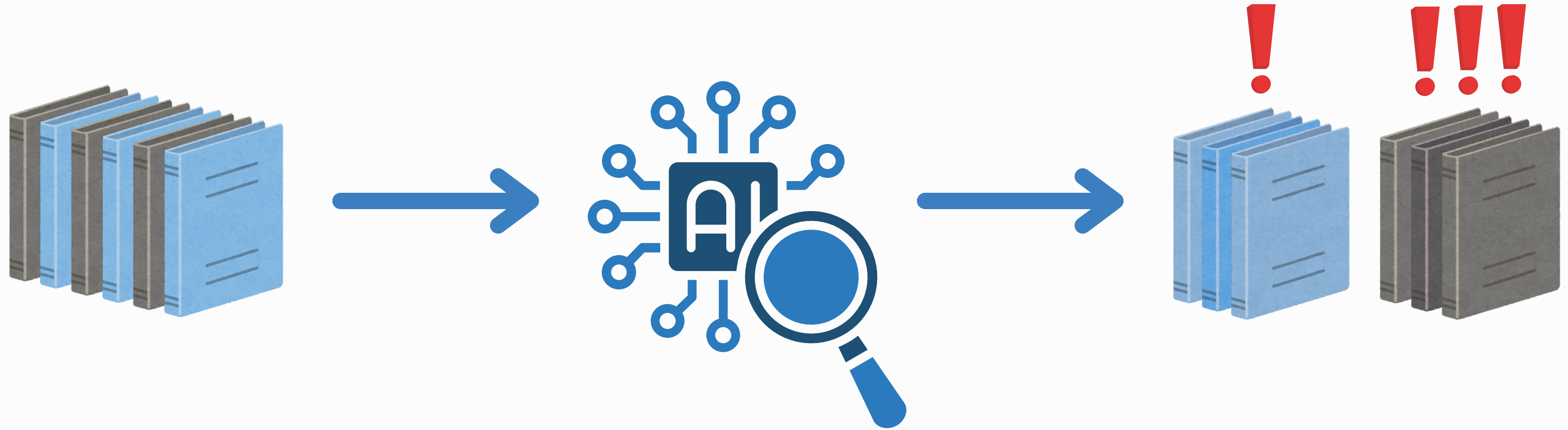
The Vision

Where Do We Go From Here?

The Vision

Where Do We Go From Here?

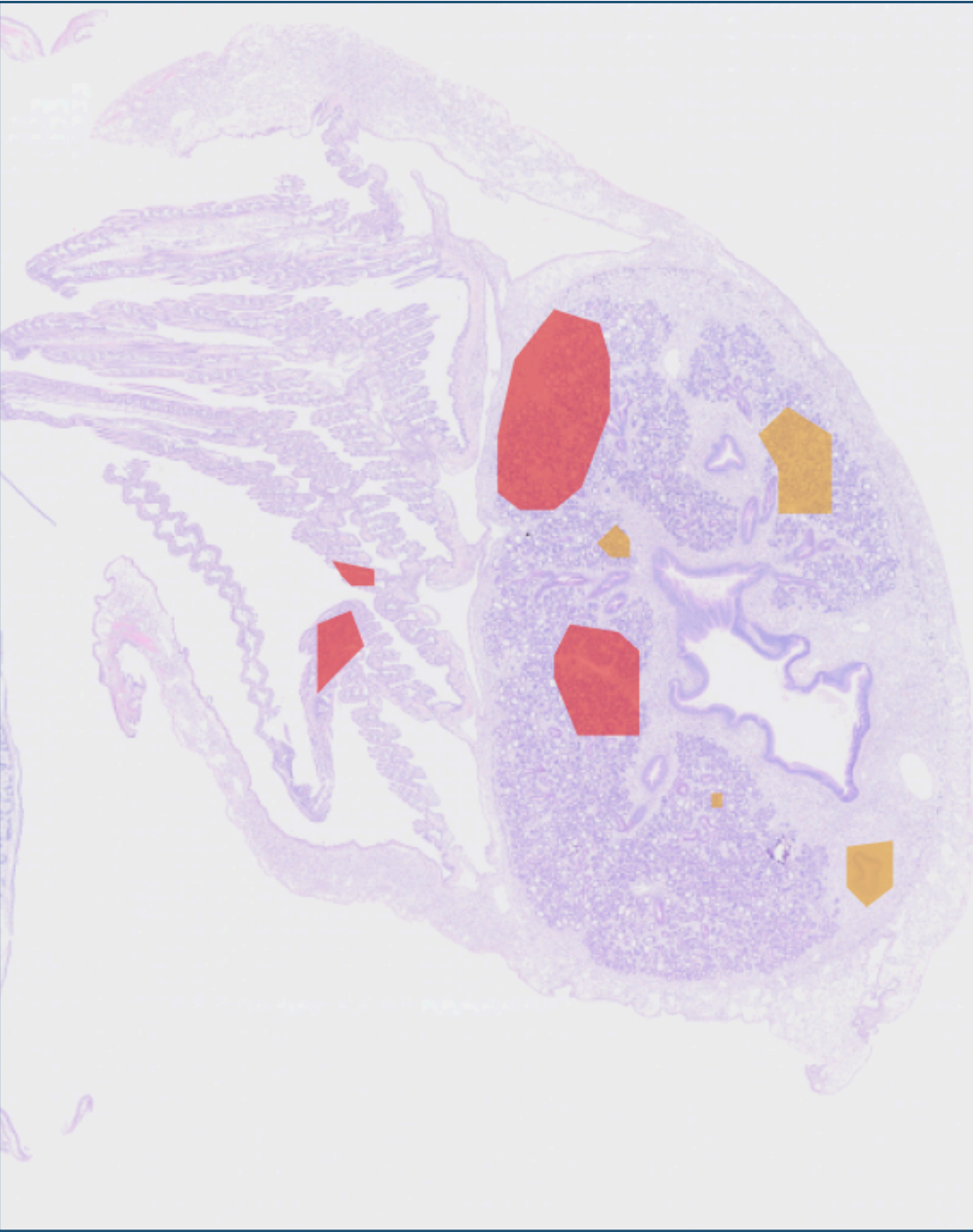
Path 1: The Triage Assistant



The Vision

Where Do We Go From Here?

Path 2: The Interactive Digital Co-Pilot



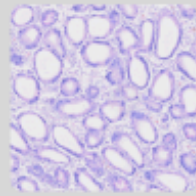
Case ID: PEI-2025-042
Client: John Smith Farms
Date Scanned: 2025-11-18

AI Summary

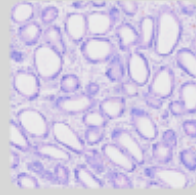
MSX DETECTED

Widespread hemocyte infiltration and multiple sporocyst clusters detected in the digestive diverticula.

Key Findings



Sporocyst Cluster 1
98% Confidence



Sporocyst Cluster 2
92% Confidence



High Inflammation Level
82% Confidence

Show me an example of sporulation

Highlight all areas with severe inflammation

Show me an ex

Write Something...

↑

The Vision

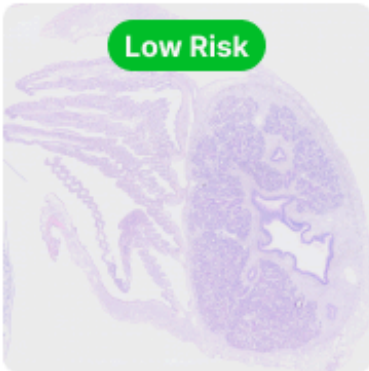
Where Do We Go From Here?

Path 3: A Screening Tool for Growers

OysterHealth Portal

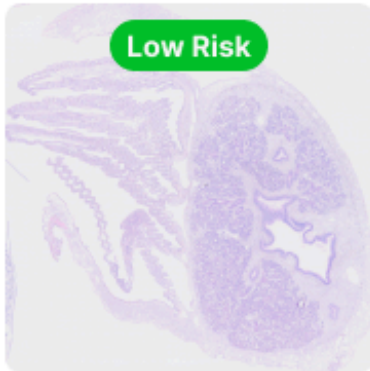
 East River Oyster Farms

East River Oyster Farms – Weekly Screening



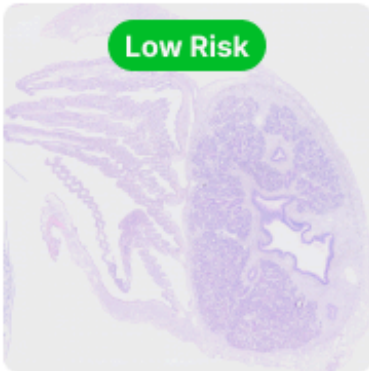
Low Risk

Sample #001



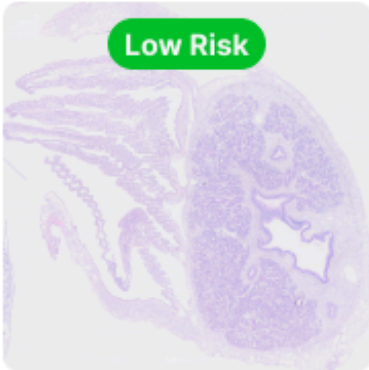
Low Risk

Sample #002



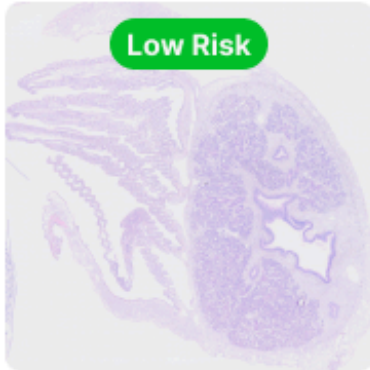
Low Risk

Sample #003



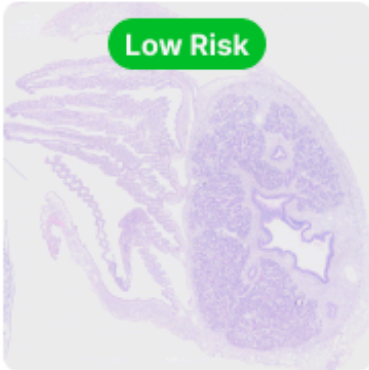
Low Risk

Sample #004



Low Risk

Sample #005



Low Risk

Sample #006

Sample ID: #493
Location: Charlottetown, PEI



ANOMALLY DETECTED

The AI System has detected cellular irregularities consistent with high inflammation.
Further analysis by a pathologist is recommended.

[Request Expert Review](#)

NOTE:
This system is for screening purposes only.
A definitive diagnosis can only be provided by a certified pathologist.

Conclusion & Takeaways

Foundation Established

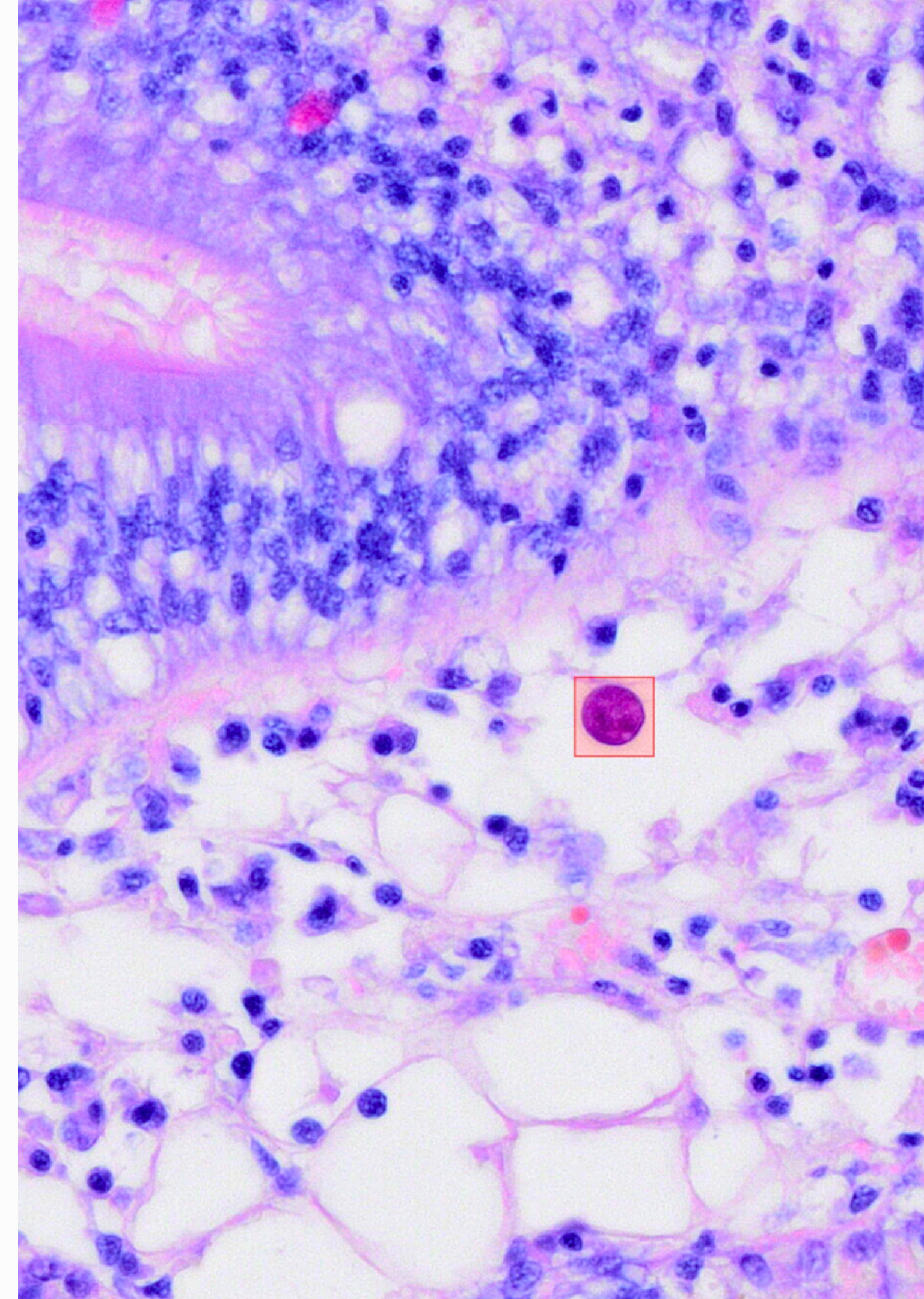
We've successfully engineered a working prototype of an AI-assisted system that is already being used to build our knowledge base.

New Capabilities

This technology opens up three distinct paths for development: Triage, Interactive Analysis, or Screening.

Future Direction

The next step is to explore these options to determine which implementation offers the most value to the diagnostic workflow.



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Thank You



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